Old deep-seated landslides also present a range of hydrological and geotechnical conditions that can favour the occurrence of new landslides. This path-dependency of landslides (Temme et al., 2020) is something important when assessing the potential of landslide disturbances in a landscape. Old landslides can sometimes reactivate. Although such transition from relatively stable conditions to important failure is relatively rare (Lacroix et al., 2020), this hazard is something not to be ignored. In a landscape, old deep-seated landslides represent a potential constraint.

Therefore, the inventory map of the old and recent deep-seated landslides that we have identified is here considered as a landslide susceptibility product that locates the places that are the most prone to be affected by ground deformations. A caveat for the delineation of the deep-seated landslides, especially for the old ones, is that, due to the interpretation of the morphology of the hillslopes, identifying the exact limits of the failed mass may not be easy and that the delineated areas must be considered with caution.

Old and recent landslides are shown on the landslide susceptibility maps for the four study areas that have been produced and are submitted together with the present report.

- **Inventory 2 – used as a dependent variable for the regional shallow landside initiation susceptibility and hazard assessment:**

For the susceptibility of the initiation of new shallow landslides, we used the 2,544 recent shallow landslide observations we made in previous research in western Rwanda. These landslides were identified by Depicker et al. (2020a) through the visual analysis of Google Earth images over the 2000-2019 period. As explained in Step 1, all recent landslides have their timing of initiation well constrained between the timing of two satellite images; which is mandatory for the hazard assessment (Step 3). The recent shallow landslides were not triggered by seismic activity and we can expect that in most situation rainfall is at their origin (Monsieurs et al., 2018; Depicker et al., 2020a, 2020b, Dewitte et al., 2021), although the role of rock weathering cannot be excluded for some landslides occurring in isolation (Dille et al., 2019). Since we look at the susceptibility associated with the occurrence of new landslides, we used the source area of each landslide in our assessment. More specifically, the source area is represented by one initiation point that was manually defined (e.g. Jacobs et al., 2018; Depicker et al., 2020a, 2020b). For the shallow susceptibility assessment we therefore used a dependent variable containing 2,544 landslide initiation points. In other words, each landslide is represented by 1 pixel in the analyses, either at 10 or 30 m resolution, depending on the topographic data source that is used (Table 1).

The impact zone of the landslides was not considered in the susceptibility analysis (Figure 3 b). The average area of these shallow landslides is 725 m², with 121 m² and 604 m² for the source and impact areas respectively.

We stressed earlier when explaining the shallow landslides, (see page 82) that an inventory from satellite images will always provide an underestimation. This concerns of course the inventory we use here. Since our inventory was compiled with the consistent use of Google earth images, we know that this underestimation is similar across the whole region. Therefore, the use of this inventory will not impact the reliability of the susceptibility assessment, the latter being data-driven and therefore relying of relative assessment. For the underestimation of the landsides, more details are provided in Depicker et al. (2020a). For the susceptibility assessment more details are provided in Depicker et al. (2020b)."